Artificial Intelligence

BSc Computer Science - Durham University

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1 Module Introduction

Artificial Intelligence has the module code COMP2261, and is led by Dr Yang Long.

1.1 Assessment

Title	Opening	Due	Feedback	Professor	Weight
Machine Learning	9th Oct	15th Dec	21st Jan	Dr Yang Long	50%
AI Search	15th Jan	18th Feb	18th Mar	Prof Iain Stewart	25%
Bias in AI	5th Feb	30th Mar	6th May	Dr Shauna Concannon	25%

Table 1: Assessments for Artificial Intelligence.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	14:00 - 16:00	Computer Class	TLC025	-
Tuesday	16:00 - 17:00	Lecture	W103	Dr Yang Long
Friday	15:00 - 16:00	Lecture	W103	Dr Yang Long

Table 2: Michaelmas Term Timetable for Artificial Intelligence.

2 Machine Learning

This section is taught by Dr Yang Long. The recommended reading for this topic is:

- Mathematics for Machine Learning (Deisenroth, M.P., Faisal, A.A., and Ong, C.S., 2020).
- Mastering Machine Learning with scikit-learn (Hackeling, G., 2017).

2.1 Introduction to Machine Learning

2.1.1 Module Information

This module is organised around a direct connection between teaching and practical work. The lectures introduce the concepts, mathematics, assumptions, and experimental consequences needed for that week's practical. The practical then produces the evidence used in the corresponding coursework report. This can be seen in Table 3.

Stage	What you do	What you keep
Lecture	Understand the main question, mechanisms, assumptions, and key equations.	Notes in your own words and questions that still need an answer.
Practical	Implement, test, compare, visualise, and discuss the method.	A notebook that runs from top to bottom with outputs included.
Weekly Report	Select the strongest evidence and explain why the result occured.	A concise report that links theory, method, result, and interpretation.
Portfolio	Revise the nine reports and connect ideas across the module.	A coherent record of technical progress and developing judgement.

Table 3: The learning and coursework model.

For more information about expectations for evidence, see (Long and Durham University, 2026, pp.3-4).

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2.1.2 What is Machine Learning?

Definition 2.1.2.1: Arthur Samuel said in 1959 that machine learning is the

> "field of study that gives computers the ability to learn without being explicitly programmed."

The field arose from the question – can a computer learn on its own in order to perform a specific task, rather than relying on a programmer explicitly setting up rules for it? Similarly, Tom Mitchell defined machine learning in 1997 as

> "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E."

From this, we can define a well-defined learning task as $< T, P, E >$.

There are three typical types of machine learning, these are:

- **Definition 2.1.2.2**: Supervised learning seeks to learn the mapping (rules) between inputs and outputs. Labelled data (correct answers) is provided in past input and output pairs during the learning process to train the model on how it should behave for previously unseen data.
- **Definition 2.1.2.3**: Unsupervised learning seeks to discover the hidden pattern from a set of inputs. Unlabelled data is provided as the past input during the learning process. The program will then attempt to group similar instances into groups.
- **Definition 2.1.2.4**: Reinforcement learning works by providing occasional positive and negative feedback to reinforce behaviours. Good behaviours are rewarded with treats and therefore become more common whereas bad behaviours are punished and therefore become less common. This is a balance between exploration (of unchared territory) and exploitation (of current knowledge).

2.1.3 Differences between AI, ML, and DL

Definition 2.1.3.1: Artificial intelligence is defined as human intelligence being exhibited by machines.

Definition 2.1.3.2: In simple terms, machine learning can be defined as an approach to achieve artificial intelligence.

Definition 2.1.3.3: Deep learning is defined as a technique for achieving machine learning. As such, we can consider AI to be a subset of ML and ML to be a subset of DL.

2.2 Machine Learning Workflow and Coding Environment

2.2.1 Machine Learning Workflow

Definition 2.2.1.1: To train a model, you use data to create a model and fine-tune it to make better predictions. We can then use this predictive model to make predictions on previously unseen data in order to perform specific tasks.

The machine learning workflow is set out as shown below:

1. **Definition 2.2.1.2**: Project Framing: When setting out to create a model, we must first frame the problem. Here, we set out a clear objective that we want the model to achieve. We then define inputs and outputs and establish success metrics (e.g., prediction accuracy).
2. **Definition 2.2.1.3**: Data Preparation: The next step is to prepare the data. To do that, we identify features (inputs) and a label (output). We then carry out data cleaning on the dataset and split the data into training and test sets.
3. **Definition 2.2.1.4**: Algorithm Selection: Next, we select the algorithm we want to use for the model based on the dataset and the objective.

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4. **Definition 2.2.1.5**: Model Training: The training data is used to incrementally improve the model's performance. The progress made in this step is dependant on the quality of the work done in the previous two stages.
5. **Definition 2.2.1.6**: Model Testing: The next step is to take the model and test it against previously unseen data (the test set).
6. **Definition 2.2.1.7**: Hyperparameter Tuning: We can then attempt to improve the training process to further increase the model's accuracy.
7. **Definition 2.2.1.8**: Inference/Prediction: Finally, the model can be deployed for the task outlined in the first step.

2.2.2 Coding Environment

When programming, we traditionally set out the features as variable X and the target label as Y.

2.3 Data Understanding

Definition 2.3.0.1: A sample is the unit for which the model recieves features and, during training, a label.

We then set up a feature matrix X where each row is a different sample and each column is a new feature and y is a matrix where each row is sample's label corresponding to the sample's features in X.

Definition 2.3.0.2: Nearest-centroid classification attempts to classify the samples by setting out one centre for each species and then classifying them based on their nearest centroid. The formula for calculating this is:

$$\mu_c = \left(\frac{1}{N_c}\right) \sum_i : y^{(i)} = cx^{(i)} \tag{1}$$

For each species c, average the feature vectors belonging to that species. Then, subtract the centroid from the new example, square the differences, add them, then take the square root, as shown below.

$$d(x, \mu_c) = ||x = \mu_x||_2 \tag{2}$$

To predict a sample's species, compute the distances from each species' centroid and choose the class with the smallest distance.

$$\hat{y} = arg\ min_c||x - \mu_c||_2 \tag{3}$$

2.4 Data Preparation Workflow

Definition 2.4.0.1: To prepare data for machine learning, we use the data preparation workflow which follows the steps as below:

1. Read and inspect the data.
2. Clean and split the data into train and test sets.
3. Fit the baseline and model.
4. Test and compare the model against the baseline.
5. Scale the model and predict unseen data.

Definition 2.4.0.2: We can standardise our dataset using training-set standardisation to reduce the unbalanced impact features with large average Euclidean distances have on predictions. To do this, we use (4).

$$x' = \frac{x - \mu_{train}}{\sigma_{train}} \tag{4}$$

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2.5 Data Construction

2.5.1 Analysing the Raw Data

Definition 2.5.1.1: Data can be considered skewed to the right if the majority of the data lies to the left with a long tail to the right. To measure this, we use a measure called skewness which can be calculated with (5)

$$\text{Skewness} = \frac{3 \times (\text{Mean} - \text{Median})}{\text{Standard Deviation}} = \frac{3(\mu - Mdn)}{\sigma} \tag{5}$$

Definition 2.5.1.2: Datasets may sometimes contain capped features. This can be identified by graphing the data. If there is a spike on the maximum-value, then it is likely that the data is capped.

2.5.2 Sampling

When deciding how large of a sample to take from a population, you have to balance the size as a larger sample reduces random sampling noise, but also increases processing time.

Definition 2.5.2.1: The Kolmogorov-Smirnov test compares the distributions of two samples. It returns the maximum vertical distance between two cumulative distribution functions.

$$D = sup_x |F_{sample}(x) - F_{population}(x)| \tag{6}$$

It can then be used to calculate a p-value measuring the evidence against equal distributions. This means that if it is above 5%, then there is insufficient evidence of a difference in the distributions.

Definition 2.5.2.2: Similarly, for categorical data, you can use the Chi-square goodness of fit test which compares the expected and observed counts of each unique value.

$$\chi^2 = \sum_j \frac{(O_j - E_j)^2}{E_j} \tag{7}$$

where O_j is the observed count and E_j is the expected count.

2.5.3 Building Trustworthy Evaluation Sets

A row should only serve one role. Either as a training set (to fit parameters), a validation set (to compare choices), or as a test set (to get a final accuracy estimate).

Random splitting preserves population proportions only in expectation. A finite test set can contain too many or too few examples from an important region. Rare strata show the largest relative variation.

2.6 Data Transformation

3 Artificial Intelligence Search

This section is taught by Professor Iain Stewart. There is no recommended reading for this topic however there may be an additional video to watch after lectures to enable more content to be covered.

Watch the FAQs video.

3.1 Coding Real-World Problems for Search Tree Expansion

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.2 Uninformed Search: BFS, DFS, and Iterative Deepening

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

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3.3 General Heuristic Search: Greedy Best-First Search and A* Search

3.4 A* Search and Deriving Iterative Deepening A* Search

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3.9 General Versions of Particle Swarm Optimisation and Differential Evolution

3.10 Applying Particle Swarm Optimization to the TSP

4 Ethics and Bias in AI

This section is taught by Dr Shauna Concannon. The recommended reading is handled on a weekly basis for this topic.

4.1 Introduction to AI Ethics and Related Concepts

4.2 Societal Impacts of AI

4.3 Defining Bias and Fairness in ML

4.4 How to Build Better Systems

4.5 Value Sensitive Design

4.6 Evaluations and Audits

4.7 Bias Mitigation Strategies

4.8 Transparency and Accountability for Trustworthy Systems

4.9 Explainable AI

4.10 Future Trends and Ethical Frontiers

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Part I
Appendix

5 Bibliography

5.1 Module Professors

Module Lead: Doctor Yang Long

- Focus: Scalable AI solutions for large-scale healthcare applications.
- Email: yang.long@durham.ac.uk
- Office Hours: Monday 15:30 - 16:30 in Term 1.

Lecturer: Professor Iain Stewart

- Focus: Computational complexity, graph/group theory, and theoretical aspects artificial intelligence.
- Email: i.a.stewart@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00 in Term 2.

Lecturer: Doctor Shauna Concannon

- Focus: Computational linguistics, social interaction, and human-computer interaction.
- Email: shauna.j.concannon@durham.ac.uk
- Office Hours: Thursday 14:30 - 15:30 in Term 2.

5.2 References

5.3 List of Figures

5.4 Reading List

5.4.1 Machine Learning

Long, Y. and Durham University (2026). COMP2261_Machine_Learning_Study_and_Coursework_Guide-_with_Navigation.pdf.

Deisenroth, M.P., Faisal, A.A. and Ong, C.S. (2020). Mathematics for Machine Learning. [online] Cambridge University Press. Available at: https://mml-book.com/ [Accessed 3 Oct. 2026].

Hackeling, G. (2017). Mastering Machine Learning with scikit-learn. 2nd edn. Packt Publishing.

5.4.2 Ethics and Bias in AI

Various, will be added over time.